$$\mu_{i,r}^{(l)} = \frac{1}{|D_r|} \sum_{t \in D_r} x_i^{(l)}(t), \quad \nu_i^{(l)} = \frac{1}{\left|\mathcal{D}_{\text{base}}\right|} \sum_{t \in \mathcal{D}_{\text{base}}} x_i^{(l)}(t). \quad (1)$$

We then define the role-specific difference-in-means vector:

$$d_{i,r}^{(l)} = \mu_{i,r}^{(l)} - \nu_i^{(l)} \quad (2)$$

By computing $d_{i,r}^{(l)}$ for each $r \in R$, we obtain $|R|$ (29) distinct directions, each representing the shift in model activations specific to a given role. These vectors are informative in two ways: their *direction* indicates how the mean activations for role-specific and generic prompts diverge; their *magnitude* quantifies the extent of this difference.

We aim to assess model performance across these different directions using the test dataset $\mathcal{D}_{\text{test}}$. This evaluation allows us to measure how various directions influence the model's behaviour, particularly in terms of performance across the different splits $D_c \in \mathcal{D}_{\text{test}}$.

Using the identified directions, we apply two types of interventions: *activation addition* and *directional ablation*. These techniques allow us to manipulate the model's activations by reinforcing or suppressing specific directional components in the residual stream.

Activation Addition. Given a difference-in-means vector $d_{i,r}^{(l)} \in \mathbb{R}^{d_{\text{model}}}$ extracted from layer l, we can modulate the influence of the corresponding feature through a simple linear transformation. Specifically, we add the direction vector to the activations of a base input, shifting them toward the mean activation observed for role-enhanced inputs:

$$x^{(l)'} \leftarrow x^{(l)} + \alpha d_{i,r}^{(l)}. \quad (3)$$

here α is a scalar hyperparameter that scales the difference-in-means vector $d_{i,r}^{(l)}$, controlling the magnitude of the shift applied to the base activations $x^{(l)}$ toward the role-enhanced mean.

Notably, this operation is applied exclusively at layer l and affects all token positions, ensuring a controlled perturbation of the model's internal representations.

Directional Ablation. To investigate the role of a direction $\hat{r} \in \mathbb{R}^{d_{\text{model}}}$ in the model's computation, we apply *directional ablation*, which removes its contribution from the model's activations. This process effectively zeroes out the component of each residual stream activation x along $\hat{d}_{i,r}^{(l)}$, preventing the model from utilizing this direction:

$$x' \leftarrow x - \hat{d}_{i,r}^{(l)} \hat{d}_{i,r}^{(l)\top} x. \quad (4)$$

This operation is performed at every activation $x_i^{(l)}$, across all layers l and all token positions i, ensuring that the model no longer represents the targeted direction in its residual stream.

By applying these interventions, we can assess the functional role of specific directions in the model's representation space. We evaluate the impact when explicitly reinforced through activation addition and suppressed via directional ablation.

3.3 Evaluation Method

For each model and every role $r \in R$, we assess whether incorporating through Activation Addition the computed role-specific difference-in-means vectors $d_{i,r}^{(l)}$ yields an improvement in performance on the test dataset $\mathcal{D}_{\text{test}}$, with particular emphasis on the corresponding domain-reference split D_c. Tab. 2 presents the role-split relevance data reported in (Zheng et al., 2024).

Split	Role(s)
econ	economic researcher, economist, financial analyst
eecs	electronics technician, data scientist, electrical engineer, software engineer, web developer
law	bailiff, lawyer
math	data analyst, mathematician, statistician
medicine	nurse, doctor, physician, dentist, surgeon
natural science	geneticist, biologist, physicist, teacher, chemist, ecologist
politics	politician, sheriff, enthusiast, partisan
psychology	psychologist

Table 2: Split and associated roles, adapted from (Zheng et al., 2024)

For each test dataset $D_c \in \mathcal{D}_{\text{test}}$, we assess performance using a logit-based framework. Given a query $x_{r,i} \in D_c$, let $\mathbf{z} \in \mathbb{R}^{|\mathcal{V}|}$ denote the logits at the final token position, where $|\mathcal{V}|$ is the vocabulary size. The softmax function calculates the probability of each token $t \in \mathcal{V}$. Restricting our attention to the candidate answer tokens $\mathcal{T}_{\text{ans}} = \{t_A, t_B, t_C, t_D\}$, the predicted token is determined by

$$t^* = \arg\max_{t \in \mathcal{T}_{\text{ans}}} p(t). \quad (5)$$

The prediction $s(x_{r,i})$ is considered correct if t^* equals the correct answer token. Overall performance is computed as the mean of the individual scores (the percentage of correct answers).

We also investigate the magnitude α of these directions. One might hypothesise that increasing their magnitude would enhance the effect associated with a given role; however, such amplification may deteriorate text generation performance concurrently (Liu et al., 2023b; Scalena et al., 2024). $\mathcal{A} = \{\alpha_1 = 1, \alpha_3 = 3\}$ is the set of activation addition coefficients. We also evaluate the impact of ablating the direction entirely, a trade-off explored in the literature relating to safety mechanism in models (Wei et al., 2024; Arditi et al., 2024).

Let $\mathcal{M}$ denote the set of models under evaluation, R the set of roles, L the set of layers, and I the set of token positions. We formalise our grid evaluation procedure as follows; for each model $m \in \mathcal{M}$ and for each role $r \in R$, we define the intervention grid:

$$\mathcal{G} = \{(l, i, \alpha) \mid l \in L_{80\%},\, i \in I,\, \alpha \in \mathcal{A}\}, \tag{6}$$

where $L_{80\%} \subset L$ denotes the first 80% of layers to avoid interference from unembedding directions, ensuring that the selected direction is not overly proximate to the unembedding directions, following the work done by (Arditi et al., 2024). Intuitively, one could increase performance by encouraging the model to generate correct answers by aligning its activations with the unembedding directions corresponding to 'A', 'B', 'C', or 'D', which would directly incentivize the model to output the correct tokens. However, we do not consider the latter 20% of layers since our approach focuses on higher-level features focusing on the role and avoids token-level manipulation focusing on selecting the correct answer. For each tuple $(l, i, \alpha) \in \mathcal{G}$, we modify the residual stream activation $x^{(l)}$ via Eq. 3. Let $s_{m,r}^{(l,i,\alpha)}$ denote the performance (e.g., the proportion of correct answers) on the domain-specific test split $D_c \in \mathcal{D}_{\text{test}}$ after applying the intervention specified by (l, i, α). For each fixed α ($\alpha_1 = 1$ and $\alpha_3 = 3$), we select the optimal intervention tuple for role r on model m by computing

$$(l^*(\alpha), i^*(\alpha)) = \arg \max_{(l,i) \in L_{80\%} \times I} s_{m,r}^{(l,i,\alpha)}. \tag{7}$$

Optimality of a direction $d_{i,r}^{(l)}$ is defined as its ability to improve performance in the reference domain-specific split D_c when added and either degrade or maintain it when removed, compared to the baseline. Among all directions satisfying this criterion, the one with the highest performance according to Eq. 7 is considered optimal. With $d^*(\alpha)$ identified, we evaluate its effect on all other domain-specific splits D_c for $c \not\sim r$ to measure its specificity.

Additionally, we wish to quantify the proportion of optimal directions d^* that align with the role-specific signal identified via the patch-scoping procedure. We adopt the methodology outlined in (Chen et al., 2024; Ghandeharioun et al., 2024), i.e. *patch scoping*. We prompt the model with the query "*What does ⧫ represent?*", apply the Activation Addition intervention with a coefficient $\alpha = 3$, and generate an explanation based on the resulting activations, as exemplified in Fig. 3. This procedure allows us to determine whether the introduced direction corresponds to the specific role of interest.

Patch Scoping Example (Llama 3.1 8B)

Prompt: *What does ⧫ represent?*

Model Baseline Response:
The symbol ⧫ is a Unicode character known as "Heavy Large Circle." It is often used in mathematics, particularly in geometry and algebra, to represent a circle or a set of points. However, without more context, it's difficult to provide a more specific interpretation.

Model Intervention Response (Layer 20, Position -4, Chemist direction):
The ⧫ symbol is a non-specific or *non-chemical* radical, which is a *small molecule* or functional group used to represent a *specific chemical structure* or functional group in a reaction mechanism or during synthesis.

Figure 3: Diverse interpretations of ⧫, before and after model intervention.

Finally, we perform a directional ablation on d^*:

$$x' \leftarrow x - \hat{d}^* \hat{d}^{*\top} x, \tag{8}$$

We denote the performance under ablation by $s_{m,r}^{\text{abl}}$. A performance drop relative to the non-ablated case is expected.

4 Results

Experimental setting. Our evaluation uses open-source, instruction-tuned language models, focusing on the most recent versions employed by (Arditi et al., 2024). Specifically, we analyse Meta's Llama 3 series (Dubey et al., 2024), including the 3.1 8B model and the 3.2 version at 1B and 3B parameters, as well as Google's Gemma 2 (2B and 9B) (Team et al., 2024) and Qwen (1.8B and 7B) (Bai et al., 2023). We do not consider base versions of the models (non-instruction tuned).

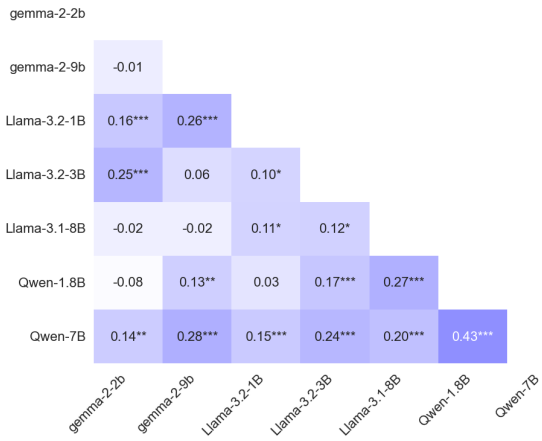

Figure 4: Spearman correlation of the percentage improvement in performance (relative to baseline) between each model after applying *activation addition*. * corresponds to p-values ≤ 0.05, ** ≤ 0.01, *** ≤ 0.001.

Our evaluation was conducted on the Cineca Leonardo supercomputer (Turisini et al., 2023). The assessment required approximately 4,500 GPU hours to process all requests, computing over 213 million inferences (29 roles r multiplied by 2457 questions, the number of layers l, coefficients a and positions i for each of the seven models $\mathcal{M}$).

RQ1: Analysis of best performing directions. Following the procedure delineated in Section 3, we apply *activation addition* with the direction $d^*(\alpha)$ for each role $r \in R$ and model $m \in \mathcal{M}$ to systematically visualise and quantify its impact on performance across $\mathcal{D}_{\text{test}}$. We compute and visualise the correlation matrix of the percentual difference relative to baseline in intervention effects across models to analyse the relationship between models. Fig. 4 shows the Pearson correlation coefficient between each model pair. Qwen-7B-Chat exhibits the highest average correlation with all other models, indicating that its behaviour under steering is consistent with other models. For this reason, we show in Tab. 3 Qwen-7B-Chat behaviour on performance for each role when steering using $d^*(\alpha)$. We display the performance scores across the eight dataset splits for all 29 roles. We report the baseline score and percentual increment over the baseline for each domain and role after applying activation addition with α of 1.0 and 3.0. The colour represents the extent of the change compared to the baseline (without intervention).

RQ2: Analysis of directions corresponding to the role. For every model $m \in \mathcal{M}$ and each direction $d^{(l)}_{i,r}$, we apply the *patch scoping* procedure (detailed in Section 3) to prompt Claude 3.5 Haiku (Anthropic, 2024a). We ask the model whether the applied direction aligns with the corresponding role $r \in R$ with the prompt shown in Fig. 5. Tab. 3 shows, among the directions that improve the baseline score on the domain-specific test split $D_c \in \mathcal{D}_{\text{test}}$, the percentage of those $d^{(l)}_{i,r}$ that Claude identifies as role-specific.

Prompt to Evaluate Patch Scoping

We are testing whether the LLM has been successfully steered from its baseline output to adopt a *{role}* perspective. Your task is to determine if the response contains relevant content to the *{role}* domain. Note that you are not assessing the quality of the response—only its relation to the *{role}* concept.

Please keep in mind: (i) The text may be repetitive or somewhat incoherent. (ii) If the response closely mirrors the baseline without introducing any distinct *{role}*-related elements, it should be considered as not aligned with the *{role}* role. Evaluate this step by step and answer the following question: Is the model being steered toward the *{role}* role?

Text to Evaluate: *{response_text}*; Baseline Reference: *{baseline_response}*.

Figure 5: Prompt for evaluating patch scoping output provided to Claude 3.5 Haiku.

RQ3: Directional ablation analysis. To evaluate whether the optimal direction $d^*(\alpha)$ plays a causal role in boosting performance on the test dataset $\mathcal{D}_{\text{test}}$, we ablate $d^*(\alpha)$ (with $\alpha = 1$) in Qwen-7B-Chat and present the resulting performance for each role $r \in R$ in Tab. 5.

5 Discussion

The effect of steering on model performances. In larger models, performance generally increases in the target and related domains and worsens or remains unchanged in domains unrelated to the role. Since we observe a strong correlation among larger models, as shown in Fig. 4, we show the full details of one model, Qwen-7B, in Tab. 3 due to space constraints[1]. For instance, we notice that the mathematician's role significantly improves Qwen-7B's performance in the domain test set, math, and the related field of EECS. Similarly, the doctor's role improves primarily medicine but also natural

[1]Full experimental results are available to reviewers in supplementary materials.